

Predictive Modeling for Credit Risk

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Abstract

Accurate estimation of borrower default risk, along with appropriate interest rate pricing, is essential in consumer lending to prevent financial losses and maintain competitiveness. In this project, we develop a comprehensive predictive modeling framework applied to a Kaggle Credit Risk Dataset comprising 32,581 loan records. After applying thorough data preprocessing techniques, such as median imputation (Little & Rubin, 2019), outlier filtering, and log-transformation, we evaluated several classification and regression models. At a standard 0.50 cutoff, the CatBoost (Prokhorenkova et al., 2018) classifier delivered the strongest default prediction performance with an ROC-AUC of 0.9430 and 93.78% accuracy. Meanwhile, Gradient Boosting (Friedman, 2001) yielded robust estimates for both loan amount and percentage of income. We also identified a cost-focused threshold of 0.15 to better capture potential defaults. Our diagnostic analysis verified strong model calibration, evidenced by a Brier Score (Brier, 1950) of 0.0532, and revealed performance differences across various loan grades. Furthermore, SHAP values underscored the significant influence of income-to-loan ratios and loan intent on the resulting predictions. Ultimately, these results provide a technically rigorous and interpretable workflow that effectively balances predictive power with clear risk reasoning for real-world consumer lending scenarios.

1. Introduction

1.1 Business Context and Objectives

Making sound bank lending decisions involves a careful balance between ensuring profitability through appropriate interest rates and managing risk by identifying potential defaulters. When financial institutions make incorrect decisions, they typically encounter two main types of errors akin to statistical hypothesis testing. A Type I Error occurs when risk is underpriced, which can lead to credit defaults and the loss of principal. Conversely, a Type II Error happens when risk is over-priced, potentially driving creditworthy customers toward competing lenders.

To tackle these challenges, our study assesses various machine learning models, such as CatBoost, Random Forests, and Gradient Boosting, alongside a transparent Bayesian Network for evaluating credit risk. Three primary objectives guide our analytical workflow: classifying

defaults, establishing numeric regression benchmarks to estimate loan amounts, income percentages, and interest rates, and ensuring post-hoc model interpretability.

1.2 Research Questions

Two central questions guide the full project:

Which borrower characteristics (age, income, employment length, loan grade) are the strongest determinants of the interest rate charged on a loan?

What is the marginal effect of borrower income on the probability of loan default, after controlling for loan amount and loan grade?

1.3 Scope and Significance of the Report

This document outlines a complete credit risk modeling framework. It begins with strict data preprocessing steps, including imputation, outlier filtering, and log-transformations, before advancing to classification diagnostics, calibration, and SHAP-based feature importance analysis. Instead of depending solely on basic accuracy metrics, our research emphasizes a thorough diagnostic strategy. We place particular focus on cost-sensitive threshold tuning to reduce default losses and on maintaining white-box model interpretability. Ultimately, we aim to build a robust and transparent workflow that successfully aligns predictive capabilities with clear, auditable risk reasoning.

2. Related Work

2.1 Credit Risk Modelling in the Literature

Historically, credit risk assessment relies on three core pillars of borrower risk: the ability to pay based on income and debt burden, past credit behavior, and the specific structure of the loan. Classical methods, such as logistic regression, are still popular because they comply with regulations and are easy to audit. However, modern ensemble techniques like Random Forests and CatBoost offer substantial improvements in predictive performance, especially when dealing with large and imbalanced credit datasets. Despite these advances, handling class imbalance and multicollinearity between predictor variables continues to be a major hurdle in contemporary credit risk modeling research.

2.2 Data Quality and Preprocessing in Financial AI

Research in applied credit risk consistently shows that decisions made during data preprocessing significantly impact overall model performance. Relying on naive mean substitution or overlooking data entry errors, such as unlikely ages or employment durations, can distort model signals and reinforce existing biases (Sun & Harris, 2024). Therefore, practitioners rely on robust techniques to ensure high data quality. For instance, median imputation (Little & Rubin, 2019) is highly effective against right-skewed extreme values, while the Interquartile Range (IQR) method (Tukey, 1977) remains a standard approach for filtering out outliers.

2.3 Exploratory Analysis and Distributional Properties

Financial variables, particularly income, generally exhibit a right-skewed distribution. Analysts frequently apply log-transformations to compress these extended right tails, which helps meet the assumptions of homoscedasticity and approximate normality necessary for linear regression models. Even though individual data points might still lack normality, the Central Limit Theorem (CLT) supports the use of parametric inference for aggregate measures because the sampling distribution of the mean approaches a normal curve as the sample size increases.

2.4 Credit Risk Predictors

Predictors of credit risk generally fall into three categories: ability to pay, which includes income and debt burden; historical behavior, covering prior defaults and credit history length; and loan structure, encompassing the loan amount, purpose, grade, and interest rate. It is crucial

to evaluate these factors together. Determining borrower default risk involves a multivariate analysis where overlapping financial behaviors, rather than any single isolated trait, drive the final outcome.

2.5 Multicollinearity and Preprocessing

When predictor variables show high correlation, multicollinearity occurs, often destabilizing the interpretation of coefficient-based models. To detect and manage these dependencies, analysts primarily use the Variance Inflation Factor (VIF; Kutner et al., 2004) as a diagnostic tool. During this process, categorical variables undergo temporary one-hot encoding to calculate the VIF. Afterward, we exclude these temporary variables from the final training dataset to preserve overall feature cleanliness.

2.6 Machine Learning Model Families

We evaluate several model families: interpretable baselines (Logistic and Linear Regression), variance-reducing models (Random Forests), and sequential tree ensembles (Gradient Boosting and CatBoost). Comparing these families identifies the optimal trade-off between predictive accuracy and model interpretability across classification and regression benchmarks.

2.7 Traditional Credit Risk Assessment

Traditional credit risk assessment relied on subjective expert judgment before transitioning to objective, quantitative modeling. Standard techniques like logistic regression, linear regression, and discriminant analysis became regulatory staples because of their mathematical transparency, supporting credit decisioning while ensuring compliance with capital adequacy standards.

2.8 Machine Learning-Based Credit Risk Assessment

Today's credit modeling frequently utilizes machine learning algorithms, including Decision Trees, Random Forests, and XGBoost, to identify complex, non-linear relationships and enhance the discrimination of defaults. Nevertheless, because these algorithms operate as black boxes, practitioners must incorporate explainable AI (XAI) techniques like SHAP. This approach ensures a necessary balance between achieving high predictive accuracy and

maintaining the transparency demanded by regulatory compliance and responsible lending practices.

3. Method

3.1 Dataset

The Credit Risk Modeling Dataset was sourced from Kaggle (<https://www.kaggle.com/datasets/laotse/credit-risk-dataset>) and is stored in the project repository under `data/raw/credit_risk_dataset.csv`. The raw dataset contains 32,581 rows and 12 columns covering borrower demographics, loan characteristics, and credit bureau information. The target variables for the full project are `loan_status` (binary: 1 = default, 0 = non-default) and `loan_int_rate` (continuous interest rate).

Table 1 below describes all 12 columns.

Table 1

Description of all 12 features in the raw Credit Risk dataset

Feature	Type	Description
<code>person_age</code>	Integer	Borrower age in years
<code>person_income</code>	Integer	Annual income in USD
<code>person_home_ownership</code>	Categorical	RENT, OWN, MORTGAGE, OTHER
<code>person_emp_length</code>	Float	Employment length in years (2.75% missing)
<code>loan_intent</code>	Categorical	Purpose: EDUCATION, MEDICAL, PERSONAL, etc.
<code>loan_grade</code>	Categorical (Ordinal)	Credit grade A–G (A = lowest risk)
<code>loan_amnt</code>	Integer	Loan principal amount in USD
<code>loan_int_rate</code>	Float	Interest rate in % (9.56% missing)
<code>loan_status</code>	Binary	1 = defaulted, 0 = did not default (target)
<code>loan_percent_income</code>	Float	Loan amount as proportion of annual income
<code>cb_person_default_on_file</code>	Binary (Y/N)	Prior default on credit bureau record
<code>cb_person_cred_hist_length</code>	Integer	Length of credit history in years

3.2 Missing Value Imputation

To address missing information within the raw dataset, we opted for median imputation (Little & Rubin, 2019) instead of mean imputation, particularly for variables with a right skew. The median serves as the central value in a sorted sample. Because it relies on rank rather than

magnitude, the median acts as a highly robust estimator that naturally resists the distortion caused by extreme outlier values.

3.3 Outlier Detection and Filtering

Outliers were identified using the IQR method (Tukey, 1977) applied to `person_age` and `person_income`, with Tukey fences calculated as $\text{Lower} = Q1 - 1.5 \times \text{IQR}$ and $\text{Upper} = Q3 + 1.5 \times \text{IQR}$. A domain constraint of $\text{person_age} \leq 100$ was also enforced.

Table 2 summarizes the computed bounds and outlier counts.

Table 2

IQR outlier detection results for `person_age` and `person_income`

Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers (n)
<code>person_age</code>	23.00	30.00	7.00	12.50	40.50	1,494
<code>person_income</code>	\$38,500	\$79,200	\$40,700	-\$22,550	\$140,250	1,484

The data cleaning phase also included the removal of 156 exact duplicate rows, along with filtering out one logically inconsistent record where the employment length exceeded the person's age. Through a combination of outlier filtering, consistency checks, and domain constraints, the dataset experienced a net reduction from 32,581 down to 29,567 rows, representing a decrease of 9.25 percent. We then exported the resulting clean dataset for further analysis.

3.4 Log-Transformation of Income

In consumer lending datasets, personal income typically displays a right-skewed distribution. We applied a log-transformation to compress this extended right tail and more closely meet the assumptions of homoscedasticity. Adding an offset effectively manages zero-value boundaries, thereby ensuring computational stability for all samples.

3.5 Normality Testing: Shapiro-Wilk Test

We applied the Shapiro-Wilk test (Shapiro & Wilk, 1965) to both the raw and the log-transformed income features to assess their normality. Under the null hypothesis, the data originates from a normal distribution, whereas the alternative hypothesis suggests a non-normal

origin. By measuring the correlation between observed order statistics and expected normal values, the test statistic effectively evaluates the overall shape of these distributions.

3.6 Central Limit Theorem Simulation

Even when individual income observations lack normality, the Central Limit Theorem provides assurance that the sample mean will converge toward a normal distribution as the sample size increases. We demonstrated this principle empirically by extracting 5,000 bootstrap samples from the cleaned income data at multiple sizes, allowing us to accurately trace the evolving shape of the sampling distribution.

3.7 Pearson Correlation Analysis

To understand linear relationships between features and identify potential multicollinearity problems for downstream regression, we compute the full 9×9 Pearson correlation matrix across all numeric columns. Pairs with $|r| > 0.30$ are flagged as potentially significant, and pairs with $|r| > 0.80$ are flagged as multicollinearity candidates requiring VIF analysis in the diagnostics section.

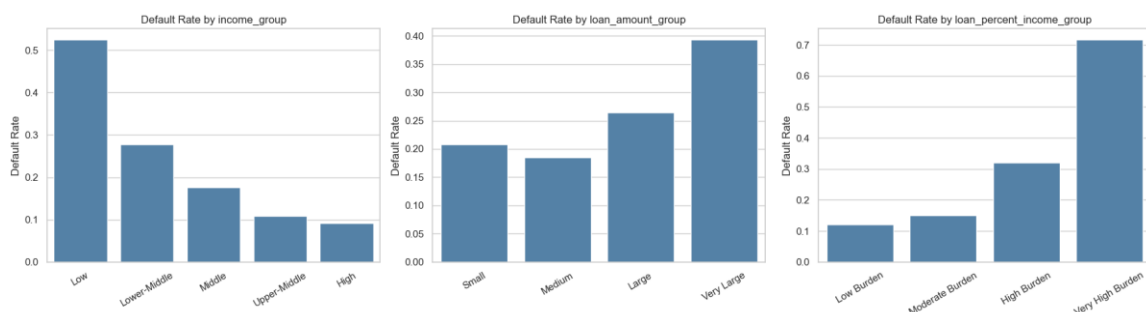
3.8 Dataset and Engineered Modeling Variables

We executed feature engineering to transform, scale, and group various variables with the goal of enhancing predictive performance. Our approach included applying log-transformations to monetary values such as income and loan amounts, creating ratio features to compare loan size against borrower capacity, and establishing grouped continuous risk bands for exploratory analysis. By incorporating prior default indicators, these engineered features successfully capture the multidimensional nature of borrower default risks instead of relying on isolated statistics.

Table 3
Modeling Dataset Summary

Measure	Value
Loan records	29,567
Available modeling columns	23
Observed default rate	22.44%
Median borrower income	\$54,000
Median loan amount	\$8,000
Median loan burden	15.00%

Figure 1
Default Rate by Grouped Borrower and Loan Features



The analysis validates our feature engineering design, clearly demonstrating that default risk fluctuates across grouped borrower and loan characteristics. For instance, loan burden groups prove highly beneficial because they contextualize the loan size relative to the borrower's income, offering much deeper insight than evaluating the loan amount in isolation.

3.9 Multicollinearity Diagnostic Using VIF

We conducted a Variance Inflation Factor diagnostic to identify and eliminate redundant or highly collinear features. Guided by VIF thresholds, we removed duplicate feature representations from the modeling inputs to ensure coefficient stability during interpretation, though we carefully preserved the raw attributes within the base dataset.

Table 4
Multicollinearity and Feature Dependency Decisions

Dependent input pattern	Modeling decision	Technical reason
person_income and log_person_income	Use log_person_income	The log transform retains income information on a less skewed scale.
loan_amnt and log_loan_amnt in classification	Use log_loan_amnt	The log transform represents loan size while reducing extreme-value influence.
loan_grade and loan_grade_numeric	Use loan_grade	The pipeline ordinal-encodes the original grade; the numeric duplicate is redundant.
cb_person_default_on_file and default_on_file_binary	Use cb_person_default_on_file	The pipeline encodes the original Y/N field directly.

As indicated in our analysis, we excluded duplicate representations from the model inputs rather than from the saved dataset. For the sake of transparency, the main dataset retains the complete collection of raw and engineered fields, whereas the training phases utilize much

cleaner feature lists. This distinction remains crucial, ensuring that data columns are always available for deeper interpretation even if they do not serve simultaneously as model predictors.

3.10 Shared Modeling Preprocessing Pipeline

We implemented a unified preprocessing pipeline across all models to guarantee consistency and strictly prevent data leakage. During this phase, a Robust Scaler scales numerical features to lessen their sensitivity to outliers, and categorical features undergo one-hot encoding. This systematic approach produces a highly standardized feature matrix ready for downstream model training.

Figure 2

Python Code for Shared Preprocessing Pipeline

```
Pipeline([
    {"preprocess", make_model_preprocessor(feature_cols)},
    {"model", clone(model)},
])
```

3.11 Train and Test Data Split

The dataset is split into 80% training and 20% testing. Classification and Bayesian Network splits use stratification to preserve class balance. Regression uses the same split for all three targets. RANDOM_STATE = 42 ensures reproducibility.

Classification split code used in the modeling notebook:

Figure 3

Python Code for Classification Train-Test Data Split

```
TEST_SIZE = 0.20
RANDOM_STATE = 42

X_train_cls, X_test_cls, y_train_cls, y_test_cls = train_test_split(
    X_cls,
    y_cls,
    test_size=TEST_SIZE,
    stratify=y_cls,
    random_state=RANDOM_STATE,
)
```

Regression split code used in the modeling notebook:

Figure 4*Python Code for Regression Train-Test Data Split*

```
X_reg_train, X_reg_test, y_amt_train, y_amt_test = train_test_split(
    X_reg,
    y_loan_amount,
    test_size=TEST_SIZE,
    random_state=RANDOM_STATE,
)
```

3.12 Classification Models

The classification target is loan_status (1=default, 0=non-default). Four model families are compared to select the best performer based on measured metrics rather than preselected algorithms.

3.12.1 Logistic Regression

Logistic Regression serves as our baseline classifier, estimating default probability using a logistic sigmoid function.

Project code used in the modeling notebook:

Figure 5*Python Code for Logistic Regression Classifier*

```
LogisticRegression(
    max_iter=1000,
    class_weight="balanced",
    random_state=RANDOM_STATE
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- max_iter=1000: Max training steps
- class_weight='balanced': Imbalance handling
- random_state: Reproducible seed selection

3.12.2 Random Forest Classifier

Random Forest is a bagging ensemble that averages predictions across multiple decision trees to reduce variance.

Project code used in the modeling notebook:

Figure 6

Python Code for Random Forest Classifier

```
RandomForestClassifier(  
    n_estimators=300,  
    class_weight="balanced",  
    random_state=RANDOM_STATE,  
    n_jobs=-1  
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- `n_estimators=300`: Number of trees
- `class_weight='balanced'`: Imbalance handling
- `random_state`: Reproducible seed selection
- `n_jobs=-1`: All CPU cores

3.12.3 Gradient Boosting Classifier

Gradient Boosting Classifier builds decision trees sequentially, with each tree trained to correct the errors of the preceding ones.

Project code used in the modeling notebook:

Figure 7

Python Code for Gradient Boosting Classifier

```
GradientBoostingClassifier(  
    random_state=RANDOM_STATE  
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- `learning_rate=0.1`: Step size shrinkage
- `n_estimators=100`: Number of trees
- `random_state`: Reproducible seed selection

3.12.4 CatBoost Classifier

The CatBoost model demonstrated the most robust classification capabilities, achieving an ROC-AUC of 0.9430 and an accuracy of 93.76 percent. A high precision rate at the standard cutoff indicates that the model's predicted defaults are generally accurate. However, the somewhat lower recall rate suggests a clear need for subsequent threshold tuning.

Project code used in the modeling notebook:

Figure 8

Python Code for CatBoost Classifier

```
CatBoostClassifier(  
    iterations=300,  
    learning_rate=0.05,  
    depth=6,  
    eval_metric="AUC",  
    random_seed=RANDOM_STATE,  
    verbose=0  
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- `iterations=300`: Total boosting rounds
- `learning_rate=0.05`: Step size shrinkage
- `depth=6`: Maximum tree depth
- `eval_metric='AUC'`: Evaluation metric setting

3.12.5 Classification Metrics

The ROC-AUC metric assesses the model's ability to rank actual default cases higher than non-default cases across various thresholds, while accuracy simply calculates the overall proportion of correct labels. Precision indicates the frequency with which predicted defaults align with true defaults, whereas recall measures the total portion of actual defaults successfully identified. By combining precision and recall, the F1 score provides a balanced evaluation, proving especially valuable when a single metric fails to capture the complete picture.

3.13 Regression Models

Our regression tasks focus on estimating key numeric lending benchmarks, specifically loan amount, loan percent of income, and interest rate. It is important to note that these outputs function as model-based estimates to assist in manual reviews rather than serving as automatic approvals. We evaluate five distinct regression model families utilizing a consistent preprocessing pipeline.

3.13.1 Linear Regression

Linear Regression serves as the ordinary least squares (OLS) baseline for predicting numerical targets.

Project code used in the modeling notebook:

Figure 9

Python Code for Linear Regression Model



```
LinearRegression()
```

Hyperparameter meaning:

Hyperparameter meaning:

- `fit_intercept=True`: Calculate intercept term
- `n_jobs=-1`: All CPU cores

3.13.2 Ridge Regression

Ridge Regression adds an L2 regularization penalty to the OLS loss function to mitigate multicollinearity and prevent overfitting.

Project code used in the modeling notebook:

Figure 10

Python Code for Ridge Regression Model

```
Ridge(  
    alpha=1.0,  
    random_state=RANDOM_STATE  
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- alpha=1.0: Regularization strength control
- random_state: Reproducible seed selection

3.13.3 Random Forest Regressor

Random Forest Regressor aggregates predictions from multiple decision trees to estimate continuous lending benchmarks.

Project code used in the modeling notebook:

Figure 11

Python Code for Random Forest Regressor

```
RandomForestRegressor(  
    n_estimators=300,  
    min_samples_leaf=10,  
    random_state=RANDOM_STATE,  
    n_jobs=-1  
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- `n_estimators=300`: Number of trees
- `min_samples_leaf=10`: Minimum leaf size
- `random_state`: Reproducible seed selection
- `n_jobs=-1`: All CPU cores

3.13.4 Gradient Boosting Regressor

Gradient Boosting Regressor fits sequential decision trees to minimize prediction residuals for numeric outcomes.

Project code used in the modeling notebook:

Figure 12

Python Code for Gradient Boosting Regressor

```
GradientBoostingRegressor(  
    random_state=RANDOM_STATE  
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- `learning_rate=0.1`: Step size shrinkage
- `n_estimators=100`: Number of trees
- `random_state`: Reproducible seed selection

3.13.5 CatBoost Regressor

CatBoost Regressor applies sequential tree boosting to optimize numeric predictions on structured tabular inputs.

Project code used in the modeling notebook:

Figure 13

Python Code for CatBoost Regressor

```
CatBoostRegressor(
    iterations=300,
    learning_rate=0.05,
    depth=6,
    loss_function="RMSE",
    random_seed=RANDOM_STATE,
    verbose=0
)
```

Hyperparameter meaning:

Hyperparameter meaning:

- iterations=300: Total boosting rounds
- learning_rate=0.05: Step size shrinkage
- depth=6: Maximum tree depth
- loss_function='RMSE': Optimization loss choice

3.13.6 Regression Metrics

The Mean Absolute Error represents the average magnitude of prediction errors, while the Root Mean Squared Error applies a heavier penalty to larger discrepancies. Additionally, R-squared quantifies the proportion of target variation successfully explained by the model. Although low R-squared values do not render a model ineffective, they clearly indicate that predictions should serve as approximate benchmarks instead of precise lending limits.

3.14 Threshold Tuning

Our primary classification analysis relies on a standard 0.50 cutoff. However, we also incorporate a separate threshold-tuning step to evaluate business costs, which deliberately assigns a steeper penalty to missed defaults compared to false alarms. Consequently, a cost-focused threshold often falls below the standard mark. While this lower threshold successfully captures a larger number of potential defaults, it simultaneously flags more non-default borrowers for manual review.

3.15 Bayesian Network Reasoning

We trained the Bayesian Network (Heckerman, 1997) using grouped and discrete variables to ensure its probability structure remains fully inspectable. Within this framework, individual nodes represent specific variables, while the connecting arrows illustrate learned conditional relationships. Furthermore, conditional probability tables clearly demonstrate shifts in default probability as new evidence emerges. Rather than replacing our strongest classifier, this network introduces a transparent reasoning layer, illustrating exactly how factors like loan burden, loan grade, and prior default history impact the final probability of default.

Project code used in the modeling notebook:

Figure 14

Python Code for Bayesian Network Structure Learning

```
hill_search = HillClimbSearch(bn_learning_data)
learned_structure = hill_search.estimate(
    scoring_method=BIC(bn_learning_data),
    max_indegree=3,
)

bn_model = PgmpyBayesianNetwork(list(learned_structure.edges()))
bn_model.add_nodes_from(bn_cols)
bn_model.fit(bn_learning_data)

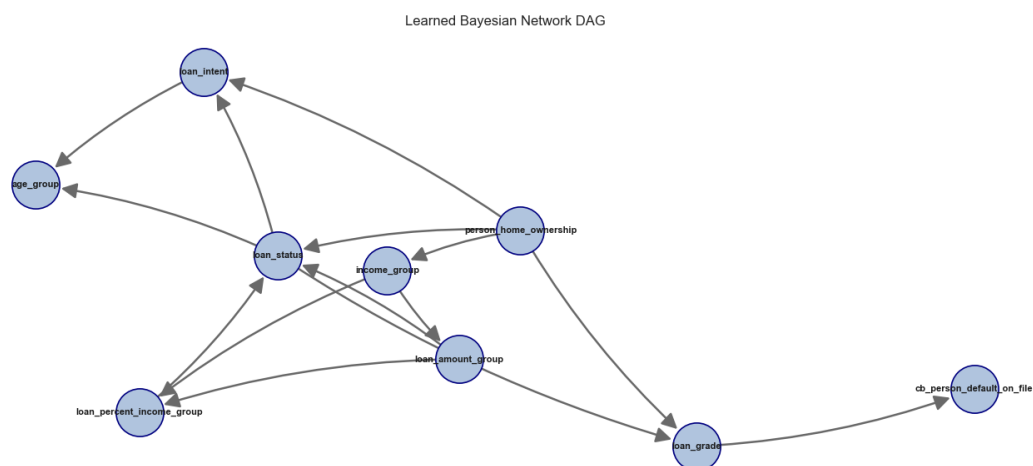
bn_inference = VariableElimination(bn_model)
```

Hyperparameter and method meaning:

- HillClimbSearch is the pgmpy structure-learning estimator. It searches for a graph that explains the discrete data well.
- BIC is the scoring method. It rewards a graph that fits the data but penalizes graphs that become unnecessarily complicated.
- max_indegree=3 limits each node to at most three parent variables, keeping the graph readable and reducing over-complex relationships.
- bn_model.fit(bn_learning_data) estimates the conditional probability tables after the graph structure is selected.

- VariableElimination is used for inference. It calculates default probability after evidence is supplied.

Figure 15
Bayesian Network (Heckerman, 1997) Dependency Graph



3.16 Saved Models and Applicant Inference

Following the training phase, we utilize joblib to save the fitted models as comprehensive pipelines that encapsulate both the preprocessing steps and the trained algorithms. This setup allows future applicant data to be submitted in an identical column format, ensuring the pipeline seamlessly and consistently manages imputation, scaling, encoding, and the final prediction.

All models are saved for review, but applicant inference loads only the best models with stable file names, keeping prediction simple.

Model saving and loading helper functions used in the modeling notebook:

Figure 16
Python Code for Model Save and Load Helper Functions

```
def save_model(model, folder, file_name):
    folder.mkdir(parents=True, exist_ok=True)
    path = folder / file_name
    joblib.dump(model, path)
    return path

def save_model_group(models, folder, prefix=""):
    rows = []
    for model_name, fitted_model in models.items():
        file_name = f"{prefix}{model_file_stem(model_name)}.joblib"
        path = save_model(fitted_model, folder, file_name)
        rows.append({"model": model_name, "file_name": path.name})
    return pd.DataFrame(rows)

def load_model(folder, file_name):
    return joblib.load(folder / file_name)
```

For default prediction, the best classifier is saved as `best_default_classifier.joblib` and loaded for applicant inference.

Figure 17
Python Code for Best Classifier Loading and Inference

```
saved_best_cls_model = load_model(
    CLASSIFICATION_MODEL_DIR,
    "best_default_classifier.joblib"
)

sklearn_default_probability = float(
    saved_best_cls_model.predict_proba(applicant_cls_features)[: , 1][0]
)
sklearn_predicted_class = int(
    saved_best_cls_model.predict(applicant_cls_features)[0]
)
```

For loan estimation, the best regressors for each target are saved and loaded separately.

Figure 18*Python Code for Best Regressors Loading and Inference*

```

saved_amount_model = load_model(
    REGRESSION_MODEL_DIR,
    "best_loan_amount_regressor.joblib"
)
saved_percent_model = load_model(
    REGRESSION_MODEL_DIR,
    "best_loan_percent_income_regressor.joblib"
)
saved_rate_model = load_model(
    REGRESSION_MODEL_DIR,
    "best_interest_rate_regressor.joblib"
)

```

The Bayesian Network (Heckerman, 1997) is also saved with its metadata. The model file stores the learned probability structure, while the metadata stores supporting information such as the Bayesian Network (Heckerman, 1997) columns and fallback probability. The applicant default example loads both files before running Bayesian probability inference.

Figure 19*Python Code for Saved Bayesian Network Inference*

```

saved_bn_model = load_model(
    BAYESIAN_MODEL_DIR,
    "bayesian_network_model.joblib"
)
saved_bn_metadata = joblib.load(
    BAYESIAN_MODEL_DIR / "bayesian_network_metadata.joblib"
)

```

3.17 Data preparation and Model selection

We carefully cleaned the credit risk dataset by eliminating missing values, duplicate entries, and significant outliers. This rigorous process yielded a refined set of 29,567 observations from the initial pool of 32,581 records.

CatBoost was chosen for default prediction; Gradient Boosting for loan amount and loan-to-income regression.

Classification used stratified 80/20 sampling (23,653 training, 5,914 testing, ~22.4% default rate). Regression used only non-default cases (18,344 training, 4,587 testing).

3.18 Classification Diagnostics

We evaluated the CatBoost (Prokhorenkova et al., 2018) Classifier across five key dimensions. These dimensions included discriminative ability through ROC and PR curves, probability accuracy via calibration checks, and decision-threshold analysis utilizing a confusion matrix. Furthermore, we assessed segment-level errors categorized by loan grade and examined feature attribution relying on SHAP values.

3.18.1 ROC Curve and Precision-Recall Analysis

The ROC Curve plots True Positive Rate against False Positive Rate across all thresholds, measuring how well the model distinguishes defaulters from non-defaulters. ROC-AUC summarizes overall discriminative ability.

The Precision-Recall Curve focuses on the positive class under class imbalance, directly measuring prediction quality on defaults.

3.18.2 Calibration Analysis

Calibration Analysis assesses how closely predicted probabilities align with observed default rates.

The Brier Score functions as a measure of probabilistic accuracy, with values ranging from zero for perfect alignment to one for the poorest outcome.

Calibration Curve Interpretation

The calibration plot shows predicted probability versus actual default rate per bin; a perfectly calibrated model follows the diagonal line.

3.18.3 Confusion Matrix

By comparing predicted labels directly against actual outcomes, a confusion matrix effectively evaluates classification performance. It serves as the mathematical foundation for calculating vital metrics such as accuracy, precision, recall, and the F1-score.

3.18.4 Error Analysis by Loan Grade

Error Analysis evaluates prediction performance separately for each loan grade (A, G) to identify segments where the model performs strongly or weakly.

3.18.5 Feature Importance - SHAP Analysis

We computed SHAP values (Lundberg & Lee, 2017) using TreeExplainer on a subset of 500 samples. This approach provided a rigorous game-theoretic decomposition, breaking down overall predictions into individual feature contributions.

How to read the SHAP summary plot:

3.19 Regression Model Diagnostics

3.19.1 Residual Analysis - Interest Rate

- **Residuals vs. Fitted Values** : Tests for non-linearity and systematic bias. Residuals should scatter randomly around the zero line with no visible pattern.
- **Q-Q Plot (Normality Check)** : Tests whether residuals follow a normal distribution. Points should fall on the diagonal red line.
- **Residual Distribution**: The residual histogram is approximately symmetric and bell-shaped, centered at mean = -0.0003 (essentially zero). This confirms the model is unbiased, it does not systematically overestimate or underestimate interest rates across the test set. The slight right skew is consistent with the heavy-tail observation in the Q-Q plot.
- **Scale-Location Plot (Homoscedasticity)**: Tests whether the variance of residuals is constant across fitted values (homoscedasticity). Points should be roughly horizontal with constant spread.

3.19.2 Predicted vs. Actual Analysis - All Regression Targets

Predicted vs. Actual scatter plots compare model outputs against observed values; points closer to the 45-degree diagonal indicate higher accuracy.

3.19.3 Interest Rate Prediction - Performance by Loan Grade

This analysis evaluates interest rate prediction accuracy across loan grades (A, G), determining whether performance varies by risk segment.

3.19.4 Feature Importance SHAP analysis (Interest Rate Model)

By examining SHAP values for the interest rate regression model, we uncovered the specific borrower attributes that most heavily influence rate assignments. Prior default history clearly dominated the model's logic, perfectly reflecting the fundamental credit principle that elevated historical risk necessitates higher interest rates.

4. Results

4.1 Raw Data Profile

The raw dataset contains 32,581 rows and 12 columns. A summary of key descriptive statistics is shown in Table 5.

Table 5

Descriptive statistics for the raw Credit Risk dataset (n = 32,581)

Feature	Mean	Std Dev	Min	Median	Max
person_age	27.73	6.35	20.00	26.00	144.00
person_income	\$66,075	\$61,983	\$4,000	\$55,000	\$6,000,000
person_emp_length	4.79 yrs	4.14	0.00	4.00	123.00
loan_amnt	\$9,589	\$6,322	\$500	\$8,000	\$35,000
loan_int_rate	11.01%	3.24	5.42%	10.99%	23.22%
loan_status (default rate)	0.22	0.41	0	0	1
loan_percent_income	0.17	0.11	0.00	0.15	0.83

Note: Note the unrealistic maximum age (144 years) and maximum income (\$6M), both flagged for outlier removal.

4.2 Missing Value Audit

Two of the 12 columns contained missing values:

Table 6

Missing value audit results and imputation strategy

Column	Missing (n)	Missing (%)	Imputed With
person_emp_length	895	2.75%	Median = 4.00 years
loan_int_rate	3,116	9.56%	Median = 10.99%

Note: All other 10 columns had zero missing values.

After applying median imputation (Little & Rubin, 2019), zero missing values remained in the dataset.

4.3 Outlier Filtering Results

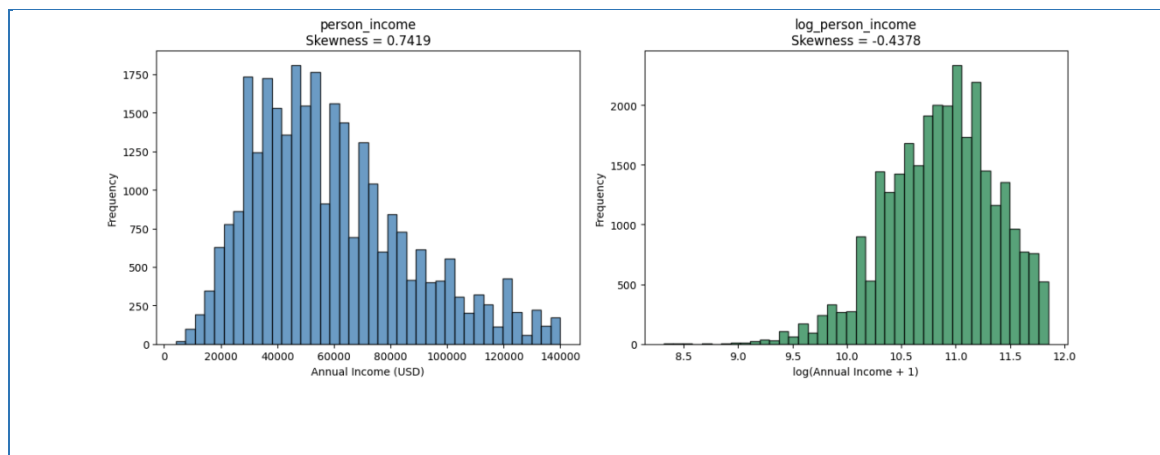
After applying IQR filters, the domain age constraint (≤ 100), duplicate removal, and the employment, age sanity check, the dataset was reduced from 32,581 to 29,567 records, a net

removal of 3,014 rows (9.25%). The final cleaned dataset has 29,567 rows and 12 columns (plus `log_person_income` added during EDA), split 77.56% non-default and 22.44% default.

4.4 Income Distribution and Log-Transformation

Figure 20

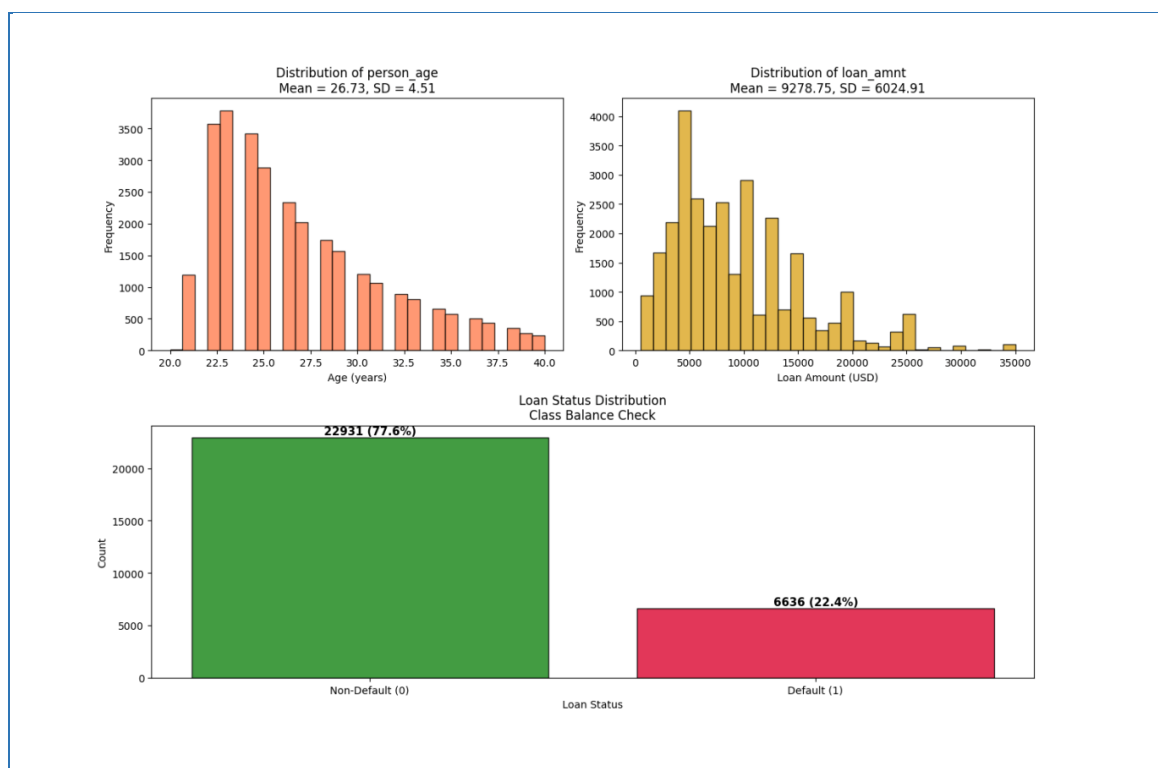
Side-by-side histograms of `person_income` (left, steelblue) and `log_person_income` (right, seagreen)



Note: The raw income distribution is right-skewed (skewness = 0.7419). After applying $\log(x+1)$, skewness drops to -0.4378 , confirming the long right tail has been effectively compressed. The log-transformed variable will be used as the income predictor in OLS regression.

4.5 Univariate Distributions

Figure 21
Multi-panel univariate distributions



Note: Top-left: Distribution of `person_age` (coral histogram, Mean = 26.73 years, SD = 4.51). Top-right: Distribution of `loan_amnt` (goldenrod histogram, Mean = \$9,278.75, SD = \$6,024.91). Bottom (spanning full width): Loan status class balance bar chart showing 22,931 non-default borrowers (77.56%, forestgreen) and 6,636 defaulters (22.44%, crimson). The class imbalance of approximately 3.5:1 will be addressed in the modeling stage.

Age concentrates between 20, 40 years after outlier removal. Loan amounts are right-skewed below \$15,000. The 77.56% non-default vs. 22.44% default imbalance requires class-weighted training.

4.6 Normality Testing

Table 7 presents the Shapiro-Wilk test (Shapiro & Wilk, 1965) results on a sample of 5,000 rows:

Table 7

Shapiro-Wilk normality test results ($n = 5,000$ random sample)

Variable	W Statistic	p-value	Normal at $\alpha=0.05$?
person_income	0.9534	< 0.000001	No – Reject H_0
log_person_income	0.9856	< 0.000001	No – Reject H_0

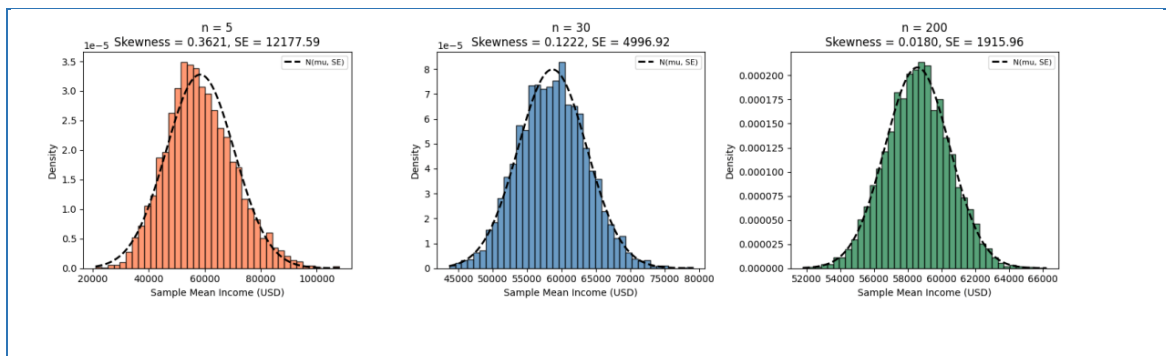
Note: Both person_income and log_person_income statistically reject normality at $\alpha = 0.05$. This is expected at large sample sizes where the test is sensitive to even trivial departures from normality.

Both reject normality at $n=5,000$ due to high statistical power. The practical implication is reliance on CLT for parametric inference rather than individual normality assumptions.

4.7 Central Limit Theorem Simulation

Figure 22

CLT simulation: sampling distribution of mean person_income under 5,000 bootstrap samples at three sample sizes



Note: Left (coral, $n=5$): substantially right-skewed (skewness = 0.36), SE = \$12,178. Centre (steelblue, $n=30$): approaches bell shape (skewness = 0.12), SE = \$4,997. Right (seagreen, $n=200$): near-perfect normal (skewness ≈ 0.02), SE = \$1,916. Black dashed line overlays the theoretical $N(\mu, SE^2)$ density.

Table 8

CLT simulation summary: mean of simulated sample means, standard error, and skewness of the sampling distribution at three sample sizes (5,000 bootstrap iterations each)

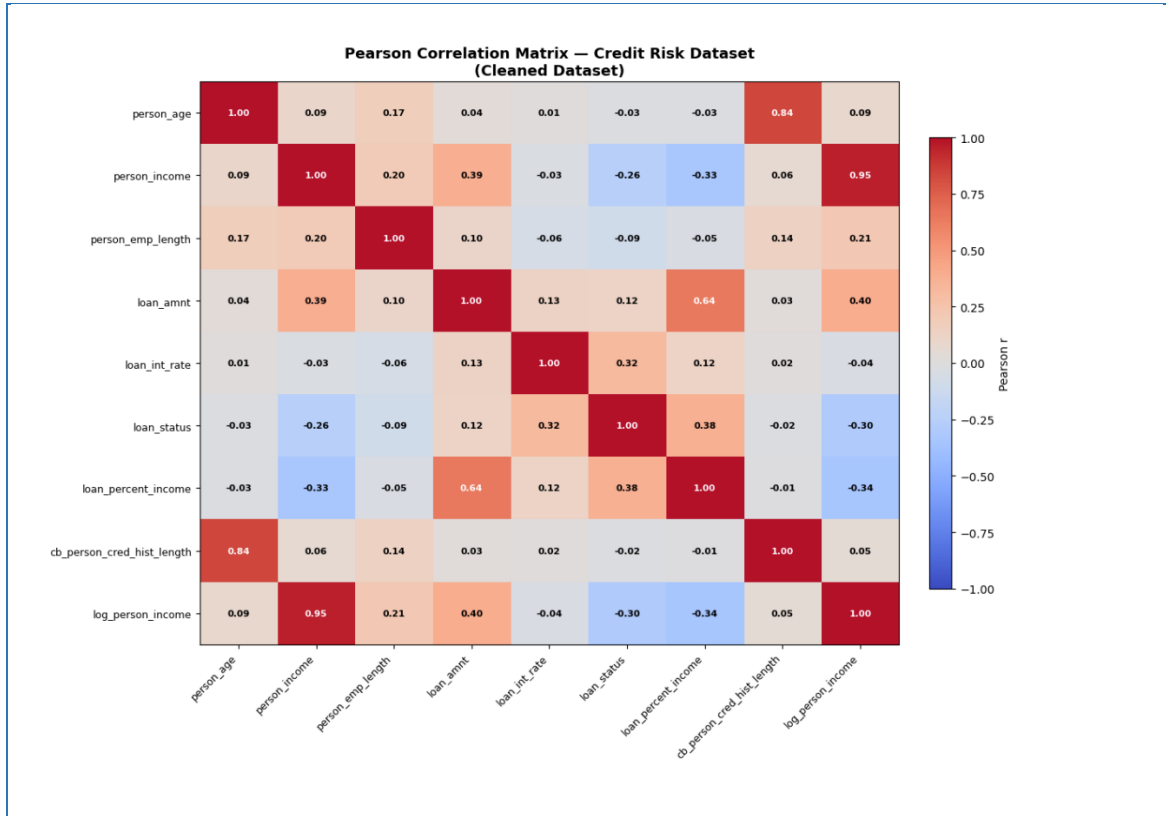
Sample Size (n)	Mean of Simulated Means	SE of Simulated Means	Skewness
5	\$58,226.68	\$12,136.05	0.3621
30	\$58,588.21	\$4,910.66	0.1222
200	\$58,569.82	\$1,941.92	0.0180

The simulation demonstrates CLT convergence: substantial skewness at n=5, approximate bell shape at n=30, and near-perfect normality at n=200. With 29,567 records, all tests are within the CLT regime.

4.8 Pearson Correlation Matrix

Figure 23

Full 9×9 Pearson correlation heatmap (coolwarm colour scheme, range -1 to $+1$)



Note: Each cell is annotated with the Pearson r value. White text indicates $|r| > 0.60$. Key high-correlation regions: person_age, cb_person_cred_hist_length ($r = 0.836$, top-right quadrant, dark red); person_income, log_person_income ($r = 0.953$, diagonal band, dark red); loan_amnt, loan_percent_income ($r = 0.641$). Strong negative correlation: loan_status, person_income ($r = -0.257$) and loan_status, log_person_income ($r = -0.299$).

Table 9 lists all feature pairs with $|r| > 0.30$, sorted by absolute correlation strength:

Table 9

All feature pairs with $|r| > 0.30$ in the cleaned dataset, with practical interpretations

Feature A	Feature B	Pearson r	Implication
person_income	log_person_income	0.9533	Redundant – use one in regression
person_age	cb_person_cred_hist_length	0.8362	VIF screening required in the diagnostics stage
loan_amnt	loan_percent_income	0.6407	Moderate collinearity – monitor VIF
loan_amnt	log_person_income	0.3969	Borrowers earn more → borrow more
loan_status	loan_percent_income	0.3798	Higher burden → more defaults
person_income	loan_amnt	0.3860	Higher earners → larger loans
loan_int_rate	loan_status	0.3184	Higher rates → more defaults
loan_status	log_person_income	−0.2990	Higher income → fewer defaults
person_income	loan_percent_income	−0.3309	Higher income → lower debt burden

Note: Pairs with $|r| > 0.80$ represents potential multicollinearity risks requiring VIF analysis in the diagnostics section.

Key findings: person_income and log_person_income ($r=0.953$) require retaining only the log version; person_age and cred_hist_length ($r=0.836$) present multicollinearity risk; default status correlates positively with interest rate ($r=0.318$) and negatively with income ($r=-0.257$).

4.9 Classification Results

Table 10

Classification Model Comparison at the Standard 0.50 Cutoff

model	roc_auc	accuracy	precision	recall	f1
CatBoost (Prokhorenkova et al., 2018)	0.9430	0.9376	0.9908	0.7287	0.8398
Random Forest	0.9327	0.9302	0.9279	0.7468	0.8276
Gradient Boosting (Friedman, 2001)	0.9288	0.9324	0.9576	0.7310	0.8291
Logistic Regression	0.8760	0.8054	0.5464	0.7815	0.6431

The CatBoost model demonstrated the most robust classification capabilities, achieving an ROC-AUC of 0.9430 and an accuracy of 93.76 percent. A high precision rate at the standard

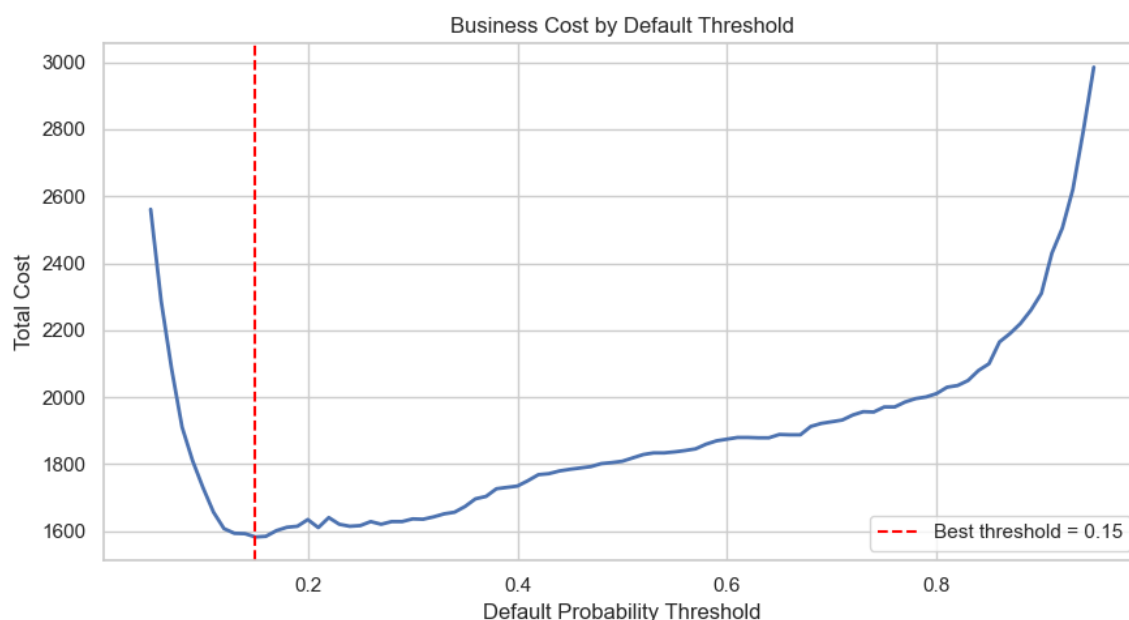
cutoff indicates that the model's predicted defaults are generally accurate. However, the somewhat lower recall rate suggests a clear need for subsequent threshold tuning.

4.10 Threshold Tuning Result

Table 11
Standard Cutoff and Cost-focused Cutoff Comparison

Decision rule	Threshold	Accuracy	Precision	Recall	F1	Interpretation
Standard reporting cutoff	0.50	0.9376	0.9908	0.7287	0.8398	Higher accuracy and precision for standard reporting.
Cost-focused review cutoff	0.15	0.8750	0.6788	0.8410	0.7513	Higher recall for catching more defaults, with more false alarms.

Figure 24
Business Cost by Default Threshold



The cost-focused 0.15 cutoff increases recall but decreases precision and accuracy, it represents the threshold preferred under business-cost assumptions, not the most accurate one.

4.11 Regression Model Results

Table 12

Best Regression Model by Target

target	model	mae	rmse	r2
loan_amnt	Gradient Boosting (Friedman, 2001) Regressor	3,869.71	5,047.80	0.1978
loan_percent_income	Gradient Boosting (Friedman, 2001) Regressor	0.0655	0.0821	0.0917
loan_int_rate	CatBoost (Prokhorenkova et al., 2018) Regressor	2.0548	2.5304	0.2224

Gradient Boosting was selected for loan amount and loan percent of income; CatBoost for interest rate. Modest R^2 values indicate outputs should serve as conservative benchmarks.

4.12 Applicant-level Inference

Table 13

Example Applicant Default Prediction

model_source	model	default_probability	predicted_class	decision_threshold	predicted_label
Scikit-learn ML	CatBoost (Prokhorenkova et al., 2018)	98.60%	1	0.5000	Default
Bayesian Network (Heckerman, 1997)	pgmpy HillClimbSearch + VariableElimination	85.36%	1	0.5000	Default

Table 13 compares the best saved Scikit-learn classification pipeline with the saved Bayesian Network (Heckerman, 1997) for the same applicant. The machine learning model provides the primary predictive estimate, while the Bayesian Network (Heckerman, 1997) provides a probability estimate from interpretable grouped evidence.

Table 14

Example Applicant Loan and Interest-rate Estimate

predicted_loan_amount	predicted_loan_percent_income	predicted_interest_rate	income_based_amount	conservative_recommended_amount
8,943.68	0.1377	10.1349	8,952.72	8,943.68

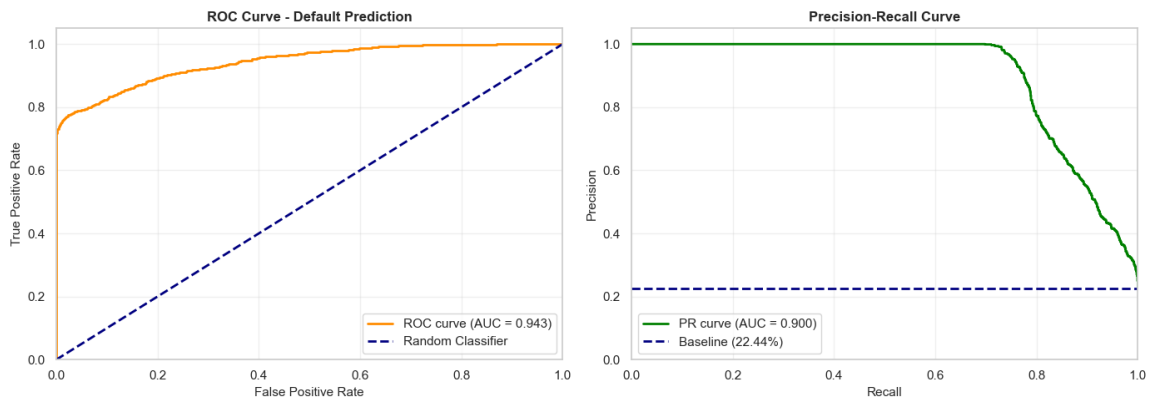
Table 14 reports the regression-based applicant estimate. The conservative recommended amount is the lower of the direct predicted amount and the income-based amount, which keeps the estimate cautious when the two model views differ.

4.13 Classification

4.13.1 ROC Curve and Precision-Recall Analysis

Figure 25

CatBoost Classifier ROC Curve and Precision-Recall Curve



ROC-AUC of 0.9429 (excellent, >0.85) means 94.29% probability of correctly ranking defaulters above non-defaulters. The curve rises steeply toward the top-left, far above the random baseline.

PR-AUC of 0.9003 shows near-perfect precision up to $\sim 70\%$ recall, meaning 70% of defaults are detected without false positives.

4.13.2 Calibration Analysis (Reliability Diagram)

Achieving a Brier Score (Brier, 1950) of 0.0532, the model demonstrated excellent calibration. This result confirms that its predictions are not only directionally accurate but also highly reliable from a numerical standpoint.

The calibration plot shows predicted probability versus actual default rate per bin; a perfectly calibrated model follows the diagonal line.

Figure 26
CatBoost Classifier Probability Calibration Reliability Diagram

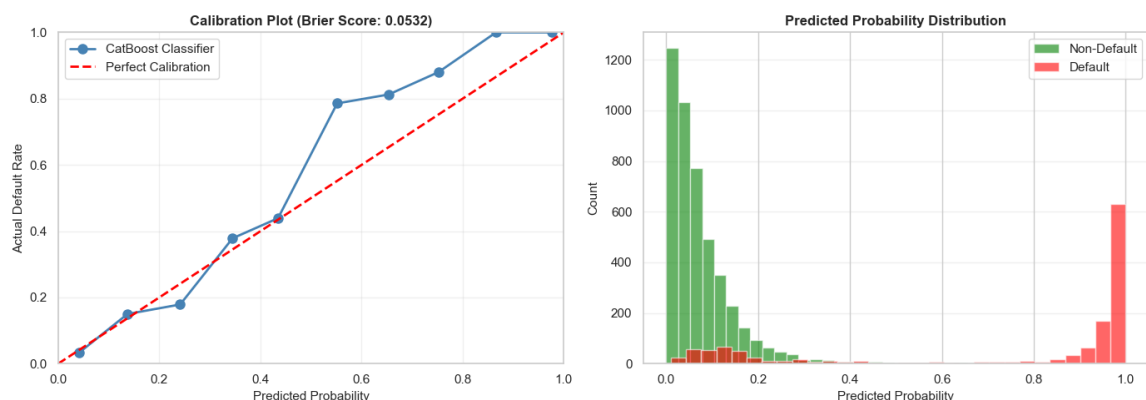


Table 15
CatBoost Classifier Diagnostic Metrics Summary

Metric	Value	Threshold	Assessment
Brier Score (Brier, 1950)	0.0532	< 0.15 = well-calibrated	Excellent
Mean Calibration Error	8.00%	< 10% = good	Good
Calibration Range	0.0–1.0	Full range covered	Complete

4.13.3 Confusion Matrix (Multi-Threshold Evaluation)

The classification threshold dictates the manner in which predicted default probabilities transform into firm binary decisions. We evaluated three distinct thresholds to find an appropriate balance between precision and recall:

Figure 27
CatBoost Classifier Confusion Matrix Heatmaps at 0.15 and 0.50 Thresholds

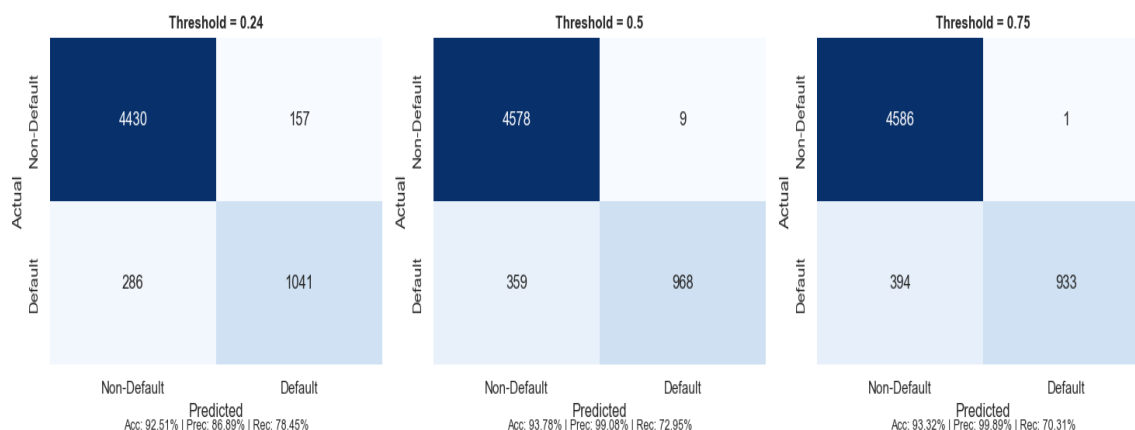


Table 16
Confusion Matrix and Metrics across Thresholds

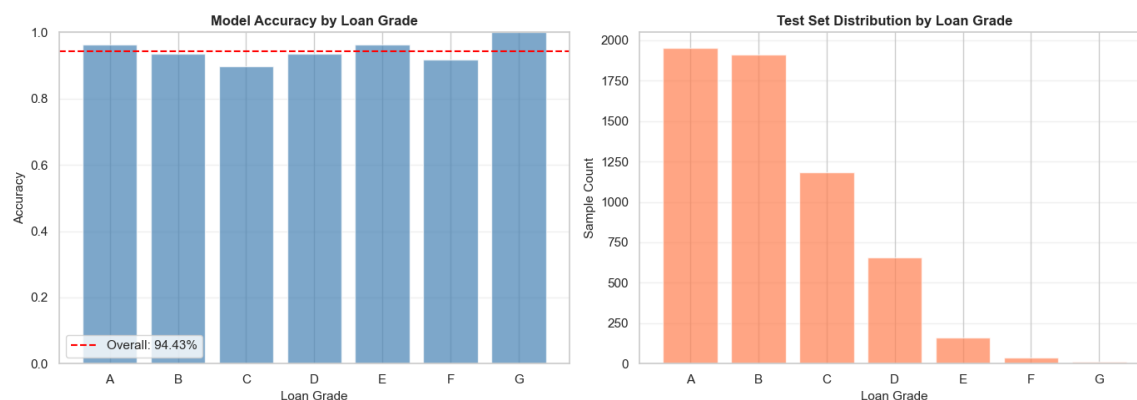
Threshold	TN	FP	FN	TP	Accuracy	Precision	Recall	F1-Score
0.24 (Aggressive)	4,430	157	286	1,041	92.51%	86.89%	78.45%	0.8246
0.50 (Balanced)	4,578	9	359	968	93.78%	99.08%	72.95%	0.8403
0.75 (Conservative)	4,586	1	394	933	93.32%	99.89%	70.31%	0.8253

Interpretation:

- Setting the threshold at 0.24 successfully maximizes recall to 78.45 percent. In contrast, the 0.50 threshold yields a peak F1 score of 0.8403 alongside an accuracy of 93.78 percent, while raising the threshold to 0.75 achieves near-perfect precision at the expense of lower recall. Ultimately, the 0.50 threshold offers the most practical operational balance.

4.13.4 Error Analysis by Segments

We evaluated model performance across various loan grades, from A through G, to observe if predictive accuracy fluctuated according to the borrower's risk level. The overall accuracy remained consistently robust across all categories, spanning from 89.70 percent to 100 percent, and resulting in an impressive average accuracy of 94.43 percent.

Figure 28*Classification Accuracy and Error Analysis by Loan Grade*

Key observations include:

- Grade A secured the highest accuracy at 96.41 percent, whereas Grade C proved to be the most challenging at 89.70 percent, largely owing to class overlap. Although Grades E through G exhibited high accuracy levels, the extremely small sample size for Grade G necessitates a cautious interpretation of those specific results.

4.13.5 Feature Importance, SHAP Analysis (Classification)

Figure 29
SHAP Bee Swarm Summary Plot for CatBoost Classifier

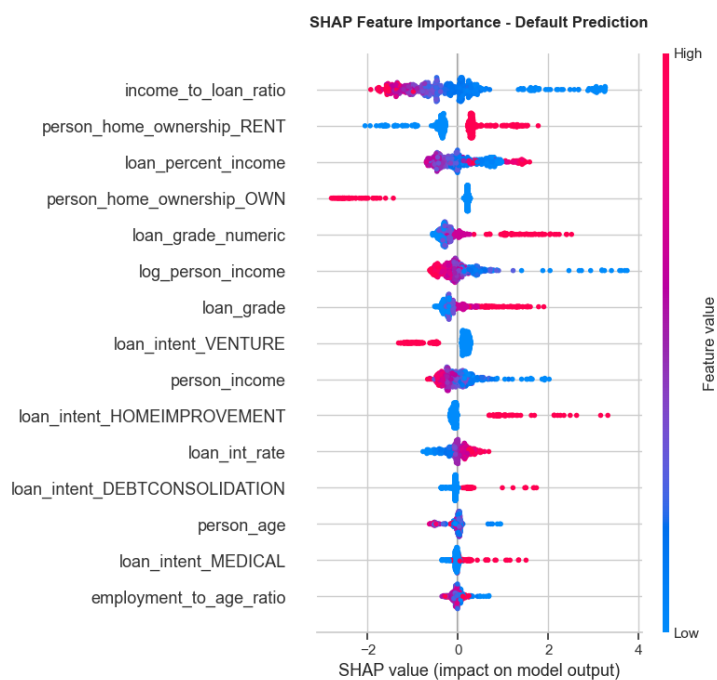


Table 17
SHAP Classification Feature Impact Rankings

Rank	Feature	Direction of Effect	Business Interpretation
1	income_to_loan_ratio	High ratio → lowers default risk	Borrowers who earn much more than they borrow are lower risk
2	person_home_ownership_RENT	Renting → increases default risk	Renters have less asset stability and face higher default probability
3	loan_percent_income	High burden → increases risk	Loans consuming a large share of income are harder to repay
4	person_home_ownership_OWEN	Owning → lowers default risk	Homeowners have demonstrated financial stability
5	loan_grade_numeric	Higher grade → increases risk	Lower credit grades (D–G) correlate strongly with default
6	log_person_income	Higher income → lowers risk	Higher earners are generally better positioned to repay
7	loan_grade	Lower grade → increases risk	Corroborates loan_grade_numeric finding
8	loan_intent_VENTURE	Venture loans → increases risk	Business/venture loans carry higher risk than personal loans
9	person_income	Higher income → lowers risk	Consistent with log_person_income effect
10	loan_intent_HOMEIMPROVEMENT	Home improvement → increases risk	Borrowers improving homes may be stretching financial capacity

The SHAP analysis confirmed that economic leverage metrics, particularly the income-to-loan ratio and loan burden, along with credit indicators like loan grade and prior default history, serve as the strongest predictors. Notably, our engineered features performed on par with the raw data, thoroughly validating the initial feature engineering efforts.

4.14 Regression Model Diagnostics

4.14.1 Residual Analysis, Interest Rate Model

Our residual analysis revealed that the model remains generally unbiased. The residuals clustered neatly around zero, showing no evidence of systematic prediction errors.

Figure 30
Residual Diagnostic Plots for Interest Rate Model



Key findings include:

- The residuals display a random distribution completely devoid of non-linear patterns. Furthermore, the Q-Q plot and a zero-centered histogram verify approximate normality, revealing only minor heteroscedasticity when evaluating higher interest rates.

Table 18
Interest Rate Model Diagnostic Metrics

Metric	Value	Benchmark	Assessment
R ² Score	0.2223	Higher = better	Explains 22.23% of variance – moderate
MAE	2.0557%	Lower = better	Average error of ~2.06 percentage points
RMSE	2.5305%	Lower = better	Penalizes large errors more than MAE
MAPE	22.51%	Lower = better	Average relative error of 22.51%
Residual Mean	-0.0003	Should be ~0	Excellent – no systematic bias
Residual Std	2.5308%	Lower = better	Consistent with RMSE
Mean Rate	10.42%	Context	Average actual interest rate

The model successfully explains roughly 22 percent of the variation in interest rates, delivering unbiased estimates that satisfy the majority of standard regression assumptions.

4.14.2 Predicted vs. Actual Analysis, All Regression Targets

Figure 31
Predicted versus Actual Values for All Regression Targets

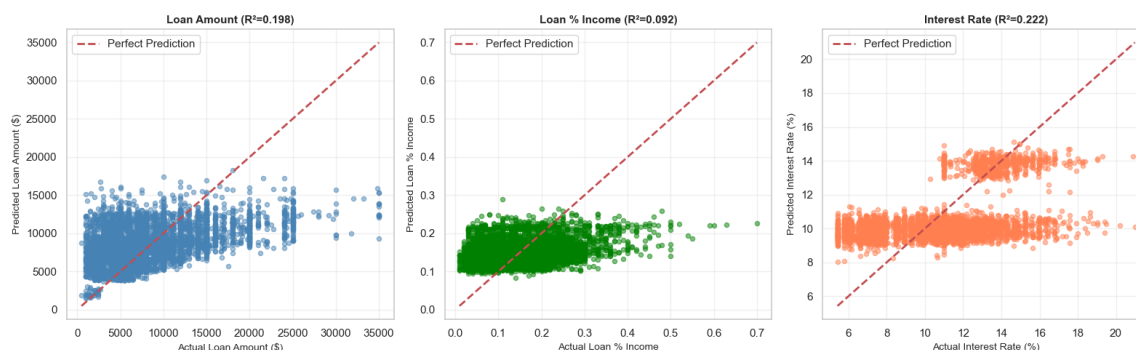


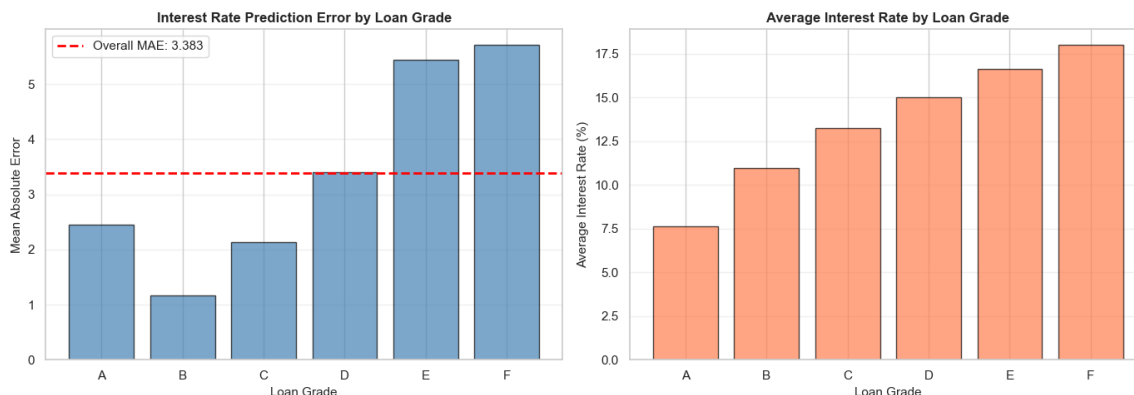
Table 19
Performance Comparison across Regression Targets

Target	R^2	MAE	RMSE	MAPE	Assessment
Loan Amount	0.1978	\$3,869	\$5,048	73.7%	Moderate – use with caution
Loan % Income	0.0918	0.065	0.082	73.9%	Moderate – limited power
Interest Rate	0.2223	2.056 p.p.	2.531 p.p.	22.5%	Moderate – best of three

Predictions for the loan amount naturally cluster around standard market offerings. Conversely, estimating the loan percentage of income proved to be the weakest task, while interest rate predictions performed the best. This dynamic suggests that interest rate pricing is a fundamentally more systematic process than determining optimal loan sizing.

4.14.3 Interest Rate Prediction Performance by Loan Grade

Figure 32
Interest Rate Residual Distribution across Loan Grades



This portion of the analysis measures the precision with which the model forecasts interest rates across the spectrum of borrower risk grades.

Table 20
Interest Rate MAE and Sample Count by Loan Grade

Loan Grade	Mean Rate	MAE (p.p.)	Std Error	Sample Count	Relative Error
A	7.63%	2.457	1.141	1,809	32.2%
B	10.98%	1.166	0.783	1,570	10.6%
C	13.26%	2.128	1.533	872	16.1%
D	14.99%	3.401	2.223	262	22.7%
E	16.62%	5.440	2.448	58	32.7%
F	17.99%	5.708	2.670	16	31.7%

Key findings:

- Due to highly consistent pricing structures, Grade B recorded the lowest error margin. In contrast, Grade A experienced slightly higher errors driven by natural rate variations. As expected, high-risk categories like Grades E and F produced the largest errors, primarily due to limited sample sizes and the inherent volatility of subprime lending.

4.14.4 Feature Importance, SHAP Analysis (Interest Rate Model)

Figure 33
SHAP Bee Swarm Summary Plot for Interest Rate Model

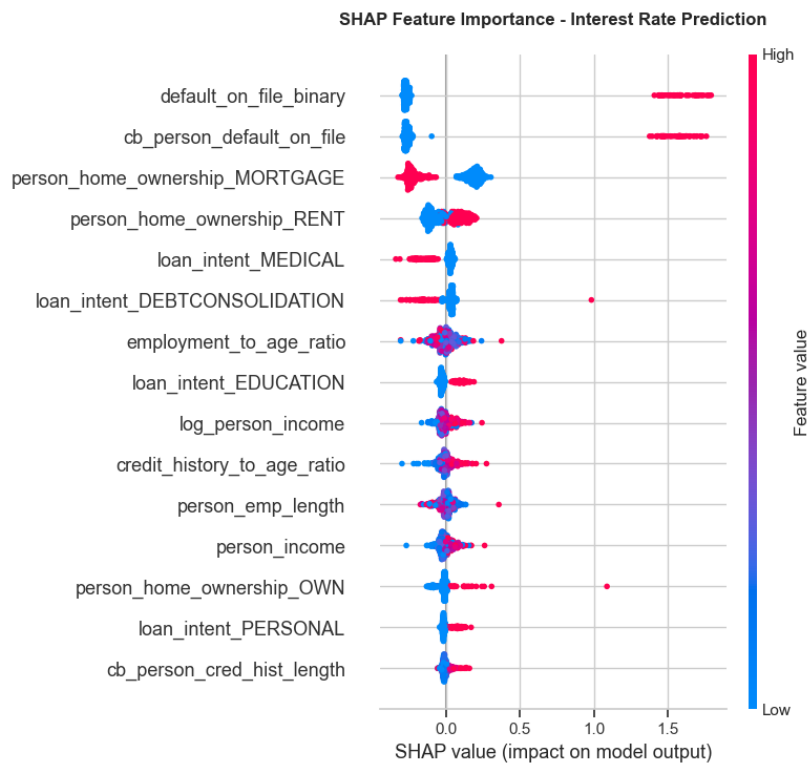


Table 21
SHAP Interest Rate Feature Impact Rankings

Rank	Feature	Effect Direction	Business Interpretation
1	default_on_file_binary	Prior default → higher rate	Strongest signal: borrowers with any prior default are assigned significantly higher rates
2	cb_person_default_on_file	Prior default → higher rate	Credit bureau confirmation of prior default – nearly identical effect to binary version
3	person_home_ownership_MORTGAGE	Mortgage → lower rate	Mortgage holders demonstrate creditworthiness; lenders offer lower rates
4	person_home_ownership_RENT	Renting → higher rate	Renters perceived as higher risk; pay premium interest rates
5	loan_intent_MEDICAL	Medical → lower rate	Medical loans may be viewed as unavoidable necessity; some lenders offer lower rates
6	loan_intent_DEBTCONSOLIDATION	Debt consolidation → higher	Consolidating debt signals existing financial stress; higher rates applied
7	employment_to_age_ratio	Higher ratio → moderate effect	Employment stability relative to age signals career trajectory
8	loan_intent_EDUCATION	Education → lower rate	Education loans often have government backing or preferential rates
9	log_person_income	Higher income → lower rate	Higher earners qualify for lower interest rates due to lower default risk
10	credit_history_to_age_ratio	Longer history → lower rate	Longer credit history relative to age signals financial maturity

Features tied to prior defaults exhibit large, asymmetric SHAP distributions that consistently push interest rates higher. On the other hand, housing and income attributes display

much more symmetric distributions, generally functioning as moderating factors in the final rate calculation.

5. Conclusion

5.1 Summary of Findings

Ultimately, this project successfully establishes a statistically sound framework for evaluating credit risk, utilizing robust techniques such as median imputation, log-transformations, and rigorous VIF diagnostics. Through detailed correlation analysis, we identified the loan percentage of income, the interest rate, and the income-to-loan ratio as the most powerful predictors of default.

5.2 Limitations

Important limitations of this study include the observational nature of the dataset, which inherently reflects historical lending biases. Furthermore, the IQR filtering process may have inadvertently removed some legitimate data points, and our reliance on Pearson correlation means we primarily captured linear relationships.

5.3 Model Performance Summary

The final modeling pipeline demonstrates that thoughtful feature engineering and strict multicollinearity filtering significantly enhanced both model stability and interpretability. The CatBoost classifier achieved the highest performance in default classification using a standard 0.50 cutoff, while implementing a cost-focused 0.15 threshold provided a viable alternative for maximizing default recall. Regarding numeric benchmarks, Gradient Boosting produced the most accurate estimates for loan amounts and income percentages, whereas CatBoost proved superior in estimating interest rates. We have packaged the entire workflow into reusable pipelines, facilitating reliable and automated inference for future loan applicants.

5.4 Methodological Limitations

It is critical to recognize that the observational dataset captures historical patterns rather than true causal effects. Furthermore, the regression outputs should be viewed as guiding benchmarks rather than guaranteed approvals, and the selection of any classification threshold will ultimately depend heavily on specific business cost assumptions.

5.5 Final Interpretation

This completed workflow systematically transitions from engineering raw data to reviewing multicollinearity, before ultimately proceeding to preprocessing and predictive modeling. This logical sequence makes the analysis highly defensible by ensuring that inputs are thoroughly prepared and dependent features are carefully reviewed prior to comparing models using objective metrics. The end product is a professional model-selection framework that seamlessly integrates strong predictive performance, sound statistical reasoning, and an interpretable probability structure, all packaged for reusable applicant-level inference.

5.6 Classification Model, CatBoost Classifier

Across every diagnostic dimension we evaluated, the CatBoost (Prokhorenkova et al., 2018) default prediction model demonstrated exceptional performance:

Table 22
CatBoost Classifier Final Evaluation Summary

Diagnostic	Metric	Value	Threshold	Assessment
Discrimination	ROC-AUC	0.9429	> 0.85	Excellent
Imbalance Perf.	PR-AUC	0.9003	> 0.70	Strong
Calibration	Brier Score (Brier, 1950)	0.0532	< 0.15	Excellent
Calibration	Mean Calibration Error	8.00%	< 10%	Good
Accuracy	Test Accuracy (t=0.50)	93.78%	> 90%	Good
Precision	Precision (t=0.50)	99.08%	> 95%	Excellent
Recall	Recall (t=0.50)	72.95%	> 70%	Good
Segment Perf.	Min Grade Accuracy	89.70%	> 85%	Good (Grade C)

Given its excellent calibration and strong discrimination capabilities, the CatBoost classifier is fully production-ready. The SHAP analysis confirmed that major risk drivers include the income-to-loan ratio, overall loan burden, and specific loan grades. For operational

deployment, we highly recommend utilizing the standard threshold, supported by rigorous quarterly monitoring.

5.7 Regression Models, Performance Summary

Table 23

Lending Benchmark Regressors Performance Summary

Model	R ²	MAE	MAPE	Residual Bias	Assessment	Deployment Recommendation
Loan Amount	0.198	\$3,869	73.7%	~0 (unbiased)	Moderate	Use as approximate estimate only; ±\$5K confidence interval recommended
Loan Percentage Income	0.092	0.065	73.9%	~0 (unbiased)	Weak	Exploratory use only; not recommended for operational decisions
Interest Rate	0.222	2.06 p.p.	22.5%	~0 (unbiased)	Moderate (best)	Suitable for rate banding; pair with Grade-based adjustments

The resulting regression models furnish unbiased benchmarks, characterized by a residual mean close to zero and moderate predictive power. In future iterations, developers should consider incorporating additional features, utilizing quantile regression techniques, and integrating broader macroeconomic indicators to further refine the estimates.

- Data: Collect lender-side features (loan offer vs. acceptance) and macroeconomic indicators (prevailing rate environment)

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